how to prserver a variance in compressed files what is meant by preserving of variance in data compression principal component analysis and singular value decomposition how they works explain in simple and easy way

Let's break this into **very simple concepts**. You are asking three related things:

1. **What does "preserving variance" mean in compression?**
2. **How PCA preserves variance in data compression**
3. **How SVD works and its relation to PCA**

1️⃣ What does "Preserving Variance" mean?

First understand **variance**.

📊 **Variance = how much the data values vary or spread out.**

Example:

Data A:

`5, 5, 5, 5` → variance = **0** (all same)

Data B:

`1, 5, 9, 13` → variance = **high** (values spread)

So:

👉 **Variance represents the information or patterns in data.**

If we compress data and **remove variance**, we lose important information.

So:

✔ Good compression tries to
reduce size but keep the variance (important information).

This is called:

Variance preservation

Meaning:

> Keep the directions where the data varies the most.

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2 Why preserve variance in compression?

Imagine a dataset:

Height	Weight
160	55
165	60
170	65
175	70

These two variables are strongly related.

Instead of storing **2 columns**, we can store **1 new variable** that captures most of the variation.

This reduces storage but still keeps most information.

This is what **PCA does**.

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3 Principal Component Analysis (PCA) – Simple Idea

PCA finds **new axes (directions)** where data varies the most.

Instead of original variables, PCA creates:

Principal Components

PC1 → direction with **maximum variance**

PC2 → second most variance

PC3 → third most variance

Example:

Original data (2D)

```
Weight
  |
  |       *
  |    *
  |  *
  |*
  +---------------- Height
```

Data mostly lies along a diagonal line.

PCA rotates the axis:

```
PC2 |
    |
    |
    |
    +------------ PC1
```

PC1 captures **almost all variance**.

So we can keep **only PC1** and discard PC2.

✔ Result:

Data compressed

Most variance preserved.

---

4 PCA Compression Example

Original data:

```
10 features
```

After PCA:

```
10 → 3 principal components
```

Compression achieved.

But those 3 components still keep **~90–95% variance**.

So information is mostly preserved.

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5 How PCA Works (Conceptually)

Steps:

1 Standardize data

Make variables comparable.

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2 Compute covariance matrix

Shows how variables vary together.

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3 Find eigenvectors and eigenvalues

Eigenvectors → directions of new axes
Eigenvalues → amount of variance in each direction

---

4 Sort by largest eigenvalues

Highest variance components first.

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5 Keep top components

Example:

Keep first 2 components instead of 10.

✔ Compression done.

---

6 Singular Value Decomposition (SVD)

SVD is a **matrix factorization method**.

It breaks a matrix into 3 parts.

If matrix = **A**

Then:

$$A = U\Sigma V^T$$

Where:

U → left singular vectors
Σ → singular values
V → right singular vectors

---

7 Simple Idea of SVD

SVD finds **important patterns in the data matrix**.

Example:

Image compression.

Image matrix:

```
1000 x 1000 pixels
```

Using SVD:

We keep only top **k singular values**.

So instead of storing full matrix, we store:

```
U_k
Σ_k
V_k
```

This reconstructs an **approximate image**.

But most information is preserved.

---

8 Relation Between PCA and SVD

Important concept:

✔ **PCA can be computed using SVD.**

In fact:

If data matrix = **X**

SVD:

$$X = U\Sigma V^T$$

Then:

Principal components come from **V**.

Variance comes from **Σ²**.

So:

👉 PCA = SVD applied to centered data.

---

9 Why PCA/SVD preserve variance?

Because they **rank components by importance**.

Largest singular values / eigenvalues =

➡ directions with **maximum variance**

When compressing we keep:

✔ largest components
❌ discard small ones

Small components mostly represent:

- noise
- very small variation

🔟 Simple Summary

Variance

➡ amount of information in data.

Preserving variance

➡ keep important patterns while compressing.

PCA

➡ rotates data to find directions with maximum variance and keeps top components.

SVD

➡ matrix decomposition used for compression and also used to compute PCA.

✅ In one line

> PCA and SVD compress data by keeping the directions where the data varies the most, so most information (variance) is preserved.

If you want, I can also show:

✔ **A super easy exam-style answer (5–6 lines)**
✔ **A visual example of PCA & SVD for exams**
✔ **How this appears in machine learning interviews**.

in which we centralize the data

We **centralize (center) the data in PCA**, not usually in basic SVD.

Let me explain simply.

---

1 What does “centralizing data” mean?

Centralizing means:

> **Subtract the mean from each feature so the average becomes 0.**

Formula:

$$X_{centered} = X - \text{mean}(X)$$

Example:

Original data (heights):

```
160, 165, 170, 175
```

Mean:

```
167.5
```

Centered data:

```
160 - 167.5 = -7.5
165 - 167.5 = -2.5
170 - 167.5 = 2.5
175 - 167.5 = 7.5
```

Now the **mean = 0**.

---

2 Where do we centralize the data?

✅ **PCA requires centralized data.**

Before applying PCA we:

1. Compute mean of each feature
2. Subtract the mean
3. Apply PCA

Reason:

PCA works using the **covariance matrix**, and covariance assumes data is centered around **zero mean**.

---

3 What about SVD?

SVD itself **does not require centering**.

SVD simply decomposes any matrix:

$$A = U\Sigma V^T$$

So:

- **Raw matrix → SVD works**
- **Centered matrix → SVD also works**

But when we use **SVD to compute PCA**, we first **center the data**.

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4 Relationship

For PCA using SVD:

Step 1: **Center data**

$$X_{centered}$$

Step 2: Apply **SVD**

$$X_{centered} = U\Sigma V^T$$

Step 3: Use **V** to get principal components.

5️⃣ Simple rule to remember

Method	Data Centering
PCA	✅ Must center data
SVD	❌ Not required
PCA using SVD	✅ Center data first

✅ One-line exam answer

> In PCA we centralize the data by subtracting the mean of each feature so that the dataset has zero mean before computing principal components.

If you want, I can also explain **why centering is important for PCA (with an easy visual intuition)** — this is a very common exam question.

matrix decomposition means to make data in higher dimension or lower

Matrix decomposition does NOT mean making data higher or lower dimension by itself.

It means:

> **Breaking a matrix into simpler matrices that reveal structure in the data.**

Think of it like **factoring numbers**.

Example:

$$12 = 3 \times 4$$

Similarly, matrix decomposition factors a matrix into multiple matrices.

Example (SVD):

$$A = U\Sigma V^T$$

Here matrix **A** is decomposed into **U, Σ, and Vᵀ**.

What happens to dimension?

Matrix decomposition itself **does not change dimension**.

Original matrix:

$$m \times n$$

After decomposition:

- **U → m × m**
- **Σ → m × n**
- **Vᵀ → n × n**

If you multiply them back, you get the **same original matrix**.

So decomposition is mainly for:

- understanding data structure
- compression
- dimensionality reduction
- solving equations

When do we reduce dimension?

Dimension reduction happens when we **keep only important components**.

Example in **SVD compression**:

Original matrix:

$$1000 \times 1000$$

Full SVD uses all singular values.

But we keep only **top k values**:

$$A \approx U_k \Sigma_k V_k^T$$

Now data is approximated using **k components**.

This results in **lower dimensional representation**.

In PCA

Matrix decomposition helps us **find important directions**.

Then we **keep only few principal components**.

Example:

Original data:

```
10 features
```

After PCA:

```
10 → 3 features
```

This is **dimensionality reduction**.

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Simple summary

Concept	Meaning
Matrix decomposition	Break matrix into simpler matrices
Dimension change	Not automatic

Concept	Meaning
Dimensionality reduction	Happens when we keep only top components

---

✅ One-line answer (exam style):

> Matrix decomposition means factorizing a matrix into simpler matrices; dimensionality reduction occurs only when we keep the most important components from the decomposition.

If you want, I can also explain **PCA vs SVD vs Eigen decomposition in one simple diagram** (very helpful for exams).